A Safety-First Two-Stage Mental Health Support Chatbot: Robust Crisis Routing and Retrieval-Grounded Response Generation from Validated Public Datasets

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Abstract. Conversational agents can widen access to emotional support, but a single misrouted message, such as a suicidal disclosure treated as small talk, can cause real harm. We present a two-stage English mental health support chatbot whose first stage is a safety router and whose second stage is a locally hosted large language model (LLM). The router is a fine-tuned RoBERTa-base model that assigns each message to one of four routes (crisis, support, small talk, out-of-scope). It is trained only on published, independently annotated datasets: MentalChat16K, CLINC150, the C-SSRS Reddit corpus, GoEmotions, HWU64, and MultiWOZ 2.2. Crisis risk is scored per sentence, thresholded on validation data with an F_2 criterion, and combined with a narrow first-person keyword rule in a two-level safety layer. The second stage uses Qwen2.5-7B with similarity-gated retrieval over 10,967 counselor responses. A ten-configuration ablation shows that in-distribution test scores hide large failures: a fine-tuned router reaching 0.833 macro-F1 on the test split scores only 0.281 on held-out and external messages, and adding figurative-language and natural out-of-scope data raises this to 0.849 (95% CI 0.794–0.889). The selected router recalls 0.914 of crisis messages from unseen C-SSRS users with a 0.3% false-alarm rate. In the generation stage, grounding rules reduce the share of replies that attribute emotions the user never mentioned from 62.0% to 8.0% on counseling inputs ($p < 0.001$).

Keywords: Mental health chatbot · Crisis detection · Intent routing · Retrieval-augmented generation · AI safety in healthcare · Suicide risk assessment

1 Introduction

Mental health is one of the most pressing public health challenges of this decade. More than one billion people live with a mental health condition, and suicide remains a leading cause of death among people aged 15–29 [1]. Shortages of trained professionals, together with stigma and a preference for self-reliance, keep many young adults from seeking help even when services exist [2]. Conversational agents have emerged as a complement: LLM-based chatbots accounted for 45% of new mental health chatbot studies in 2024 [3], and controlled trials report reductions in depressive symptoms among college students [4,5]. The same reviews, however, identify a recurring weakness: most

chatbots cannot reliably recognise and escalate a genuine emergency [5], and purely generative systems are hard to keep grounded [3].

A common remedy is to place a classifier in front of the generator. Such classifiers are usually evaluated on a random split of the corpus they were trained on. When training data come from several sources, this protocol can hide *shortcut learning* [6,7]: a model may learn that long narrative text means "support" and short commands mean "out of scope", or that the word *kill* signals danger. These shortcuts are invisible on an in-distribution split but appear immediately in real chat ("this game is killing me", "I need a cheap hotel in the centre with free parking").

This paper builds and evaluates a two-stage chatbot with these failure modes in mind. All training and test messages are drawn from published datasets, and their original annotations are mapped to chatbot routes with explicit, reproducible rules. Our contributions are:

- a safety router trained only on six published datasets, with explicit and reproducible rules that map their original labels to four chatbot routes;
- a two-level safety layer that combines sentence-level crisis scoring with a narrow first-person keyword rule, so that explicit disclosures receive a fixed hotline response and model-only alarms receive a context-aware reply that always carries hotline information;
- a ten-configuration ablation evaluated on an in-distribution split, a perturbed split, and held-out and external test sets, which quantifies how much each added dataset closes the gap between benchmark and real-chat performance;
- an evaluation of the generation stage with the seven MentalChat16K quality criteria and a reproducible measurement of emotions attributed to the user without evidence.

2 Proposed Method

2.1 Framework Overview

Fig. 1 shows the framework. Offline, six public datasets are mapped to four routes, filtered, split by group, and used to fine-tune the router; the training portion of MentalChat16K is embedded with BGE-M3 to form the retrieval index. Online, each message passes through the router, a deterministic decision layer and, when appropriate, retrieval-grounded generation by a local LLM.

2.2 Stage 1: Router and Crisis Confidence Score

The router f_θ is RoBERTa-base [8] with a four-way classification head over $\mathcal{Y} = \{$*crisis*, *support*, *small talk*, *out-of-scope*$\}$. *Crisis* means the writer discloses their own suicidal thoughts, self-harm, a plan or an attempt; *support* covers emotional difficulties without personal suicide risk; *small talk* covers greetings, thanks, jokes and casual remarks; *out-of-scope* covers requests unrelated to wellbeing.

Because a disclosure is often one sentence inside a longer message, the crisis confidence score is computed per sentence. Let $S(x)$ be the sentences of message x with at

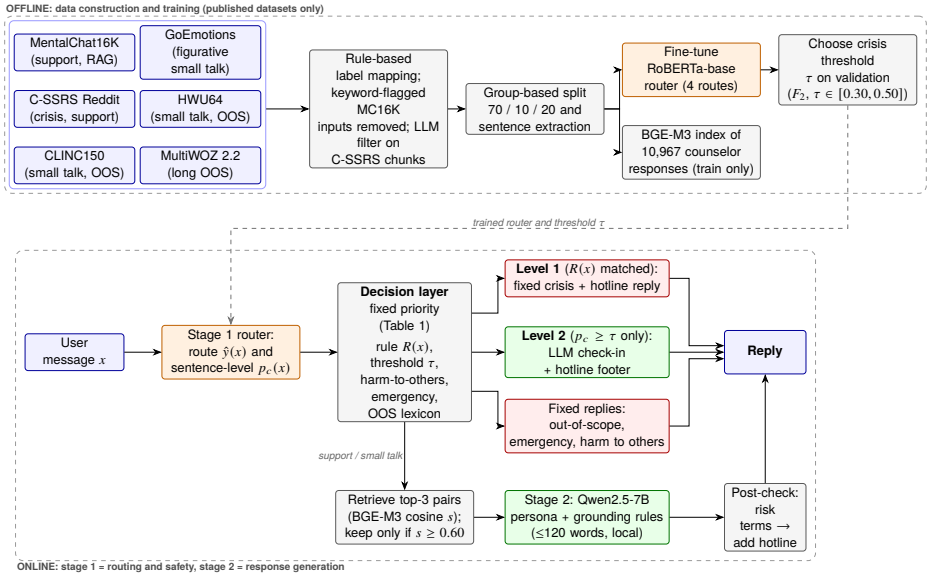


Fig. 1. Proposed two-stage framework. Offline (top): six published datasets are mapped to four routes, filtered, split by group and used to fine-tune the router and to build the retrieval index. Online (bottom): stage 1 routes the message and the decision layer applies the two-level safety logic; support and small-talk messages go to stage 2.

least two words and $P_\theta(c \mid s)$ the softmax probability of class c. Then

$$p_c(x) = \max_{s \in S(x)} P_\theta(\textit{crisis} \mid s), \qquad \hat{y}_{\neg c}(x) = \arg \max_{c \neq \textit{crisis}} P_\theta(c \mid x). \tag{1}$$

The router predicts *crisis* if $p_c(x) \geq \tau$ and $\hat{y}_{\neg c}(x)$ otherwise. The score is the model's own probability; only the threshold is tuned, on validation data, by maximising F_2 [9], which treats recall as twice as important as precision because a missed crisis is far more costly than a false alarm. Fine-tuned models produce saturated probabilities, so validation recall is flat over a wide range of thresholds and an unrestricted search is unstable (in pilot runs it returned 0.003 and 0.95). The search is therefore restricted a priori to $\tau \in [0.30, 0.50]$; the upper bound encodes the safety preference that a message the model considers at least as likely to be crisis as not is treated as crisis. The selected value is $\tau = 0.30$.

2.3 Decision Layer and Two-Level Safety

The final route is produced by a deterministic decision layer with fixed priority (Table 1). It combines the router output with four rules and uses no learned or tuned weights, so every decision can be traced to a single component. The key rule $R(x)$ matches only explicit first-person expressions ("kill myself", "I want to die", "I've been self-harming", "I'm not safe right now") and deliberately does not match mentions such as "warning signs of suicide" or "my friend self-harms".

Table 1. Priority order of the decision layer (first matching row decides).

#	Condition	Action
1	$R(x)$ matches	Level 1: fixed crisis message with hotlines (no LLM)
2	Harm-to-others rule	Fixed de-escalation message with emergency numbers
3	$p_c(x) \geq \tau$ and $R(x)$ not matched	Level 2: LLM reply that checks in directly, plus hotline footer
4	Emergency-information request	Fixed emergency and helpline information
5	$\hat{y}$ = out-of-scope, no wellbeing term	Fixed out-of-scope message
6	$\hat{y}$ = out-of-scope with wellbeing term, or support	Stage-2 support generation with retrieval
7	$\hat{y}$ = small talk	Stage-2 small-talk generation

Figurative uses of *kill*, *dead* or *die* can still receive high crisis scores. Rather than patching each idiom, level 2 lets the LLM interpret the message in context; it is instructed to ask directly whether the user is thinking about ending their life when the message could carry that meaning, which is recommended practice and does not increase risk [10]. A hotline footer is always appended at level 2, so recall is not reduced; only the form of a false alarm changes.

2.4 Stage 2: Retrieval-Grounded Response Generation

Support and small-talk messages are answered by Qwen2.5-7B-Instruct [11], served locally through Ollama [12] so no user text leaves the machine. For support messages the query is embedded with BGE-M3 [13] and compared with the inputs of 10,967 MentalChat16K training pairs. The top-3 counselor responses are inserted as tone examples only if their cosine similarity is at least 0.60, because unrelated examples led the LLM to attribute feelings the user never expressed. The prompt fixes the bot persona and adds grounding rules: respond only to what the user said, do not assume feelings or events, answer direct questions first, keep replies under 120 words. Retrieval grounding of this kind reduces unsupported content in dialogue [14,15]. If a reply contains risk-related terms, hotline information is appended. Algorithm 1 summarises inference.

3 Experiments

3.1 Datasets

All messages come from published datasets (Table 2). Table 3 gives the rule that maps each dataset's original labels to the four routes, together with the filtering applied and the number of rows used from each source.

MentalChat16K [16] contains 16,084 counseling pairs (about 6K anonymised interviews and 10K synthetic conversations); after cleaning, 15,850 remain. It has no risk labels, so we do not assign any: the 182 inputs that a broad keyword screen flags as

Algorithm 1 Two-stage inference for one user message

Input: message x; router f_θ; threshold τ; rule R; index $\mathcal{I}$; LLM G
Output: reply r

$p_c \leftarrow \max_{s \in S(x)} P_\theta(\mathit{crisis} \mid s)$; $\hat{y} \leftarrow \arg\max_{c \neq \mathit{crisis}} P_\theta(c \mid x)$
if $R(x)$ **then**
  return fixed level-1 crisis message
end if
if harm-to-others rule matches **then**
  return fixed de-escalation message
end if
if $p_c \geq \tau$ **then**
  return G(check-in prompt(x))$\oplus$ hotline footer
end if
if emergency-information request **then**
  return fixed emergency information
end if
if $\hat{y}$ = out-of-scope **and** no wellbeing term in x **then**
  return fixed out-of-scope message
end if
if $\hat{y}$ = small talk **then**
  return G(small-talk prompt(x))
end if
$E \leftarrow \{(u_i, o_i) \in \text{top-3}_{\mathcal{I}}(x) : \cos(x, u_i) \geq 0.60\}$; $r \leftarrow G$(support prompt(x, E))
if r contains risk terms **then**
  $r \leftarrow r \oplus$ hotline information
end if
return r

Table 2. Published datasets used in this study. Identifiers link to each dataset's original paper (DOI) or record; download links are given in the Data Availability statement.

Dataset	Venue	Original annotation	DOI / identifier (linked)
MentalChat16K [16]	KDD 2025	counseling input/response pairs	10.1145/3711896.3737393
C-SSRS Reddit [17]	PLOS ONE 2021	5 suicide-risk levels, psychiatrists	10.5281/zenodo.4543776
CLINC150 [19]	EMNLP 2019	150 intents + out-of-scope, crowd	10.18653/v1/D19-1131
GoEmotions [20]	ACL 2020	27 emotions, 3+ raters per comment	10.18653/v1/2020.acl-main.372
HWU64 [21]	IWSDS 2019	64 intents in 21 scenarios	arXiv:1903.05566
MultiWOZ 2.2 [22,23]	NLP4ConvAI 2020	task-oriented dialogues	10.18653/v1/2020.nlp4convai-1.13
ESConv [24]	ACL 2021	emotional-support dialogues	10.18653/v1/2021.acl-long.269

*Download links for every dataset are listed in the Data Availability statement at the end of the paper.

possibly suicidal are removed from both the router data and the retrieval index, and the crisis class comes only from C-SSRS. **C-SSRS Reddit** [17] contains posts annotated by practising psychiatrists on the Columbia Suicide Severity Rating Scale [18]. Posts are split into chunks of one to two sentences (6,212 chunks from 1,790 posts); because a post-level label does not apply to every sentence, a chunk is kept only if a few-shot Llama-3.2-3B prompt or the keyword screen confirms the post's label at the chunk level. We also found that the released text had ordinary words replaced by medical terms (e.g.,

Table 3. Mapping of original labels to the four routes, and rows used (before near-duplicate removal). In total: 321 crisis, 25,985 support, 4,629 small talk and 8,023 out-of-scope rows.

Dataset	Original label or field	Route	Rows
MentalChat16K	`input` (all counseling requests) and up to two first-person sentences extracted from it	support	21,945
	inputs flagged by a broad risk-keyword screen	removed (unlabelled)	182
	`output` (counselor response)	retrieval index	10,967
C-SSRS	*ideation*, *behavior*, *attempt*; chunk kept only if an LLM or the keyword screen confirms personal risk	crisis	321
	supportive; chunk kept only if neither flags risk	support	4,040
	indicator, uninformative labels, unconfirmed chunks	discarded	–
CLINC150	14 conversational intents (`greeting`, `thank_you`, `are_you_a_bot`, . . .)	small talk	2,097
	135 domain intents (30 utterances each) and the original `oos` class	out-of-scope	5,400
	`meaning_of_life`	ambiguous probe	30
GoEmotions	comments containing *kill/die/dead/shoot/*. . . whose rater-assigned emotions are only positive (amusement, joy, admiration, . . .), excluding any self-harm term	small talk (figurative)	321 + 29
HWU64	`general_greet`, `joke`, `praise`, `affirm`, `negate`	small talk	2,211
	all non-general scenarios (alarm, music, weather, . . . ; 20 per intent)	out-of-scope	1,123
MultiWOZ 2.2	user turns of 12–60 words (one per dialogue)	out-of-scope	1,500 + 200
ESConv	seeker's `situation` description (test only)	support	300

"like" → "Hyperactive behavior") and reverted these substitutions. **GoEmotions** [20] provides real figurative uses of violent and death-related words ("This killed me") whose non-crisis meaning is confirmed by the original raters' emotion labels. **HWU64** [21] and **CLINC150** [19] provide conversational and assistant-style utterances, and **MultiWOZ 2.2** [22,23] provides long, natural, off-topic requests. **ESConv** [24] is used only for testing.

Train/validation/test split. Each row carries a group identifier (one MentalChat16K input together with all sentences extracted from it, one C-SSRS user, one MultiWOZ dialogue, . . .), and the groups are split 70/10/20 into training, validation and test sets, so that sentences from the same conversation or the same Reddit user never appear on both sides. Rows from GoEmotions, HWU64, and MultiWOZ with a cosine similarity of at least 0.85 to any held-out message were removed from all three sets (461 rows, mostly template-like MultiWOZ requests). Table 4 lists all evaluation sets and their role. The test split and the held-out set are used only for reporting: model weights, checkpoints and the crisis threshold are chosen on the training and validation sets. The same MentalChat16K split defines the retrieval index, so stage 2 is never evaluated on conversations it can retrieve.

Test sets. Besides the in-distribution test split, we use (i) a *perturbed* split of 1,570 test messages with lower-casing, punctuation removal, character swaps and slang substitution; and (ii) a *held-out and external* set of 529 scored messages that no model sees during training: 300 ESConv situations (support; a fully external dataset), 200 MultiWOZ user turns from the original MultiWOZ test split (out-of-scope) and 29 GoEmotions figurative comments from the original GoEmotions test split (small talk). Thirty CLINC150 `meaning_of_life` utterances are reported separately as an ambiguous cat-

Table 4. Data splits and evaluation sets (after near-duplicate removal). Only the training set is used to fit models; only the validation set is used for checkpoint and threshold selection.

Set	Messages	Role
Training (70% of groups)	27,007	model fitting (213 crisis)
Validation (10% of groups)	3,776	checkpoint selection, crisis threshold τ (38 crisis)
Test split (20% of groups)	7,714	in-distribution evaluation (70 crisis, 5,198 support, 952 small talk, 1,494 out-of-scope)
Perturbed test	1,570	test-split messages with typos and slang
Held-out and external	529 + 30	sources or splits never used in training

egory. The held-out set contains no crisis messages, because no other independently annotated crisis dataset could be obtained in time; crisis recall is therefore measured on the 70 test-split crisis chunks from C-SSRS users unseen in training.

3.2 Preprocessing and Feature Engineering

Text is normalised by removing URLs, Markdown, Reddit user and subreddit references and "edit:" tails, and by collapsing whitespace. From each MentalChat16K input, up to two first-person sentences of 4–30 words are extracted (preferring the first sentence and sentences containing emotion words, excluding background statements such as age or occupation and any sentence matching the risk screen), so that the support class is not represented only by long narratives. For the frozen-encoder baselines and for retrieval, texts are encoded by BGE-M3 [13] into 1024-dimensional L2-normalised vectors. For the fine-tuned router, texts are tokenised with the RoBERTa tokenizer and truncated or padded to 128 tokens.

3.3 Machine Learning Algorithms

Logistic Regression on frozen BGE-M3 embeddings [25] (R1–R5) is used as a flat four-class classifier (R1) and as a *cascade* (R2–R5) in which a binary crisis gate is followed by a three-class router. **Fine-tuned RoBERTa-base** [8,26] (R6–R10) updates all parameters so that the representation adapts to the routing task. **Zero-shot classification** with Llama-3.2-3B [27] is a training-free baseline, using a prompt that defines the four routes and warns about figurative expressions.

3.4 Customized Ensemble Design

The system combines one learned model with deterministic rules rather than several learned classifiers. In the cascade baselines, the gate decides *crisis* if its probability exceeds its threshold, otherwise the router decides. In the deployed system, the router and the rules of Table 1 are combined by fixed priority. No combination weights are learned: with only 213 crisis training rows, a meta-learner would have little data to estimate weights for the safety-critical class, and a fixed priority guarantees that an explicit self-harm statement always receives the fixed crisis reply.

3.5 Hyperparameter Tuning

Logistic Regression uses $C = 1.0$, balanced class weights and up to 3,000 iterations; the cascade gate threshold is the value with the highest validation recall subject to at most 5% validation false alarms. RoBERTa is trained with AdamW [28] (learning rate 2×10^{-5}, weight decay 0.01, batch size 32, 3 epochs, 6% warm-up, gradient clipping at 1.0, mixed precision), cross-entropy weights proportional to $\sqrt{N/N_c}$, and, from R8 on, a cap of 4,000 support training rows. The checkpoint with the best validation macro-F1 is kept, and τ is chosen as in Section 2.2. Stage 2 uses top-3 retrieval, a 0.60 similarity gate, temperature 0.7 for support and small talk, 0.3 for level-2 check-ins and 0 for judging.

3.6 Training and Validation Protocol

All configurations use one fixed group-based split (seed 123). Training uses only the training split; the validation split is used only for checkpoint selection and thresholds; test, perturbed and held-out sets are used only for reporting. k-fold cross-validation was not used: fine-tuning takes about 4–5 minutes per configuration on a T4 GPU, and the group split already prevents the main source of optimistic bias, sentences from the same conversation appearing on both sides. Each configuration was trained once; we report 95% bootstrap confidence intervals [29] (1,000 resamples) on the held-out set.

3.7 Evaluation Metrics

Stage 1: macro-F1 over the routes present [30], crisis recall, and the false-alarm rate (share of non-crisis messages routed to crisis), plus accuracy per held-out category. The difference between test-split and held-out macro-F1 (the *gap*) measures shortcut reliance. *Stage 2:* three generators are compared on the same inputs: S1, Qwen2.5-7B with a plain counselor prompt; S2, S1 plus persona and grounding rules; S3, the full system of Algorithm 1. Inputs are 50 MentalChat16K test pairs with expert references (set A) and 60 short messages from the held-out set (set B). Each reply is scored 1–10 on the seven MentalChat16K criteria [16] by an LLM judge [31] (Qwen2.5-7B, temperature 0), which also states whether the reply assumes facts or feelings the user did not mention. As a transparent proxy for hallucinated content we report the *attributed-emotion rate*: the share of replies that, after removing fixed hotline text, contain a negative-emotion term from a fixed list of 29 stems (*overwhelm*, *stress*, *anxi*, *lonel*, ...) absent from the user's message. Paired differences are tested with the exact McNemar test. We also report BGE-M3 similarity to the expert reference and reply length.

4 Results and Discussion

4.1 Router Ablation

Table 5 reports all configurations. Three observations stand out.

(i) Test-split scores hide large failures. Fine-tuning on the core data (R5→R6) raises split macro-F1 from 0.794 to 0.820 but lowers held-out macro-F1 from 0.615 to

Table 5. Router ablation. Split: in-distribution test split (7,714 messages, 70 crisis). Held-out: 529 held-out and external messages (no crisis). FA: false-alarm rate. Gap: split macro-F1 − held-out macro-F1. Best held-out values in bold.

				Split			Held-out		Perturbed		
ID	Configuration	n_{train}	τ	F1	Rec.	FA	F1	FA	F1	Rec.	Gap
R1	LR, flat 4-class	11,037	–	.651	.986	.096	.568	.140	.742	.986	.083
R2	LR cascade (crisis gate + router)	11,037	.648	.690	.957	.042	.607	.049	.788	.914	.082
R3	+ extracted short sentences	20,802	.563	.708	.943	.047	.730	.036	.811	.900	−.022
R4	+ C-SSRS supportive as hard negatives	23,706	.421	.793	.900	.014	.615	.019	.832	.886	.178
R5	+ keyword rule $R(x)$	23,706	.421	.794	.914	.014	.615	.019	.834	.900	.179
R6	RoBERTa, same data as R5	13,444	.470	.820	.914	.005	.287	.015	.761	.900	.533
R7	+ sentence-level p_c	13,444	.300	.833	.900	.003	.281	.021	.801	.857	.552
R8	**+ rebalancing, GoEmotions, HWU64, MultiWOZ**	12,745	.300	.939	.914	.003	**.849**	**.006**	.928	.857	.090
R9	R8 without MultiWOZ	12,021	.490	.925	.943	.004	.648	.021	.929	.886	.278
R10	R8 without GoEmotions	12,514	.300	.915	.900	.005	.613	.015	.918	.900	.302
Llama-3.2-3B zero-shot (no training)			–	–	–	–	.609	.023	–	–	–

Table 6. Held-out accuracy per category.

Category (expected route)	n	R8	R9 (no MWOZ)	R10 (no GoEmo)	Llama-3B
ESConv (support)	300	.790	.840	**.900**	.760
MultiWOZ (out-of-scope)	200	**1.000**	.280	.990	.845
GoEmotions figurative (small talk)	29	**.862**	**.862**	.000	.241

0.287, and sentence-level scoring (R7) does not change this. A more powerful model fits the style of each source more closely: without other evidence, it learns that short practical problems resemble CLINC commands and that long natural requests resemble MentalChat16K narratives.

(ii) Each added dataset fixes a specific shortcut. Adding GoEmotions, HWU64, and MultiWOZ with class rebalancing (R8) raises held-out macro-F1 to 0.849 and lowers held-out false alarms to 0.6%, while keeping crisis recall at 0.914 on unseen C-SSRS users. Removing a single dataset isolates its role (Table 6): without MultiWOZ (R9), only 28.0% of long off-topic requests are recognised; without GoEmotions (R10), none of the figurative comments is routed correctly. R8 exceeds R5, R9 and R10 in every bootstrap resample (Table 7).

(iii) A zero-shot LLM is not a substitute. Llama-3.2-3B reaches 0.609 held-out macro-F1 and routes only 24.1% of figurative comments correctly; in an earlier pilot with team-written probes, it also recalled very few crisis messages, which is why it is not used as the router.

Two caveats apply. First, MultiWOZ and GoEmotions test items come from the original test splits of datasets that R8 also trains on, so they measure generalisation to unseen messages of a familiar source; only ESConv is a fully external dataset, and there R8 (0.790) is slightly below R10 (0.900), suggesting that GoEmotions comments pull some short emotional statements towards small talk. Second, the GoEmotions category is small (29 messages).

Table 7. Held-out results with 95% bootstrap confidence intervals (1,000 resamples).

	Macro-F1	False alarm	R8 minus this configuration (macro-F1)
R5	.615 [.580, .664]	.019 [.008, .032]	+.233 [.160, .292]
R7	.281 [.240, .331]	.021 [.009, .034]	+.569 [.497, .629]
R9	.648 [.594, .690]	.021 [.009, .034]	+.201 [.158, .243]
R10	.613 [.598, .626]	.015 [.006, .026]	+.235 [.181, .278]
R8	**.849 [.794, .889]**	**.006 [.000, .011]**	–

Table 8. R8 on the in-distribution test split.

Route	Precision	Recall	F1	*n*
Crisis	.711	.914	.800	70
Support	.997	.992	.995	5,198
Small talk	.960	.992	.976	952
Out-of-scope	.993	.979	.986	1,494
Macro average	.916	.969	.939	7,714

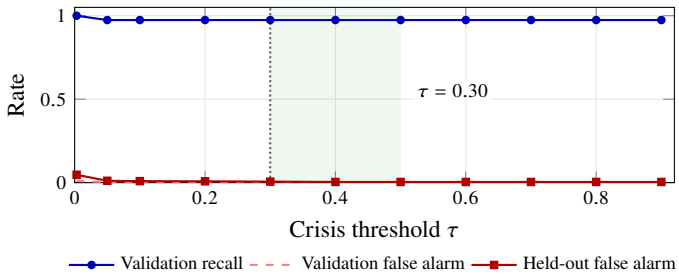


Fig. 2. Crisis recall and false-alarm rate of R8 versus threshold (model only). The shaded band is the allowed search range $[0.30, 0.50]$.

4.2 Crisis Detection, Threshold and Two-Level Safety

On the test split, R8 detects 64 of 70 crisis chunks (recall 0.914, precision 0.711) and routes 26 of 5,198 support messages to crisis (Table 8). Most of these are C-SSRS *supportive* posts that discuss suicide while comforting someone else, the hardest contrast in the data. Fig. 2 shows the effect of τ: validation recall is flat at 0.974 from 0.05 to 0.9, while held-out false alarms fall from 4.7% at $\tau = 0.003$ to 0.6% at the selected $\tau = 0.30$. All three held-out false alarms were ESConv descriptions of severe distress (e.g., "trying to find a job in this town is almost impossible ..."); none matched the explicit rule, so all were handled at level 2 with a context-aware reply and a hotline footer. The 30 ambiguous `meaning_of_life` questions were routed to out-of-scope (17) or small talk (13), never to crisis.

Table 9. Stage-2 evaluation. Criteria (1–10, LLM judge) on set A (50 MentalChat16K test inputs); average score on sets A and B (60 short held-out messages); attributed-emotion rate; similarity to the expert reference; mean length.

	Criteria (set A)							Avg.		Attr. emo. (%)			Words	
System	List.	Emp.	Safe.	Open.	Clar.	Bound.	Hol.	A	B	A	B	Sim.	A	B
S1 plain prompt	**7.92**	**8.86**	**9.00**	**9.00**	**7.78**	**9.00**	**6.78**	**8.33**	**8.14**	62.0	83.3	**.846**	163	144
S2 + grounding	7.90	8.02	**9.00**	**9.00**	7.06	**9.00**	6.04	8.00	7.97	14.0	**25.0**	.737	**40**	**29**
S3 full system	7.66	7.84	8.98	8.94	7.10	8.98	5.92	7.92	7.91	**8.0**	26.7	.750	58	37

4.3 Shortcut Analysis

A logistic-regression probe that predicts the *source* of a training text within the same route reaches 0.988 accuracy for support (majority baseline 0.538), 0.961 for small talk (0.500) and 0.905 for out-of-scope (0.728), so sources are highly distinguishable by style alone. A classifier that sees only message length reaches 0.392 macro-F1 (random: about 0.25). These diagnostics explain why each route needs several sources and why the test split rewards models that exploit style.

4.4 Evaluation of Response Generation

Table 9 compares the generators. *Grounding rules sharply reduce attributed emotions.* The plain prompt names a negative emotion the user never mentioned in 62.0% of replies to counseling inputs and 83.3% of replies to short messages; grounding rules lower this to 14.0% and 25.0% (exact McNemar, $p < 0.001$ on both sets). For example, to "The holidays always make me sad" S1 adds that the user may be "missing someone", whereas S3 reflects only what was said. Adding gated retrieval (S3) changes the rate further only on set A (14.0% to 8.0%, $p = 0.55$, not significant). *The judge favours long replies.* S1 obtains the highest average score (8.33) but its replies are about three times longer; length and score correlate at $r = 0.37$, consistent with the verbosity bias of LLM judges [31]. On safety, boundaries and open-mindedness all systems score at least 8.94. *The judge does not detect unsupported content*: it flagged none of the 330 replies, including clear cases, which is why we report the lexical proxy, an upper bound since some named emotions are appropriate reflections. *Routing errors propagate.* In set B, 13 of 60 short support messages (21.7%), mostly practical problems such as job loss, rent or COVID-19 ("I lost my job and not sure how I will pay for rent"), were routed to out-of-scope by S3 and received the fixed refusal; these lower S3's set-B score (7.65 for refusals vs. 7.99 otherwise).

4.5 Error Analysis and Comparison with Existing Work

The router's main remaining errors are (a) support messages about practical hardship routed to out-of-scope, the dominant ESConv error, because the out-of-scope sources contain many job, banking and travel requests; (b) supportive C-SSRS posts that discuss suicide on behalf of others routed to crisis; and (c) six missed crisis chunks whose risk is expressed without explicit vocabulary. Compared with existing systems, Woebot [4],

Wysa [32] and Tess [33] were evaluated on clinical outcomes rather than routing robustness, and the MentalChat16K study [16] evaluated fine-tuned LLMs on response quality without a separate safety stage. The proposed system contributes an explicit safety layer between user and generator, built from published datasets, and reports robustness beyond an in-distribution split.

5 Ethical Considerations and Limitations

The chatbot is a supportive tool, not a replacement for professional care, and it never diagnoses. Crisis information is hard-coded and never produced by the LLM. All models run locally, so messages are not sent to third parties. All datasets are used under their published terms; C-SSRS posts are anonymised by their authors.

Limitations: (1) crisis recall is measured only on C-SSRS users unseen in training (70 chunks); no independent external crisis set was available, and the held-out set contains no crisis messages. (2) Two of the three held-out sources share their origin with training data, and the figurative category is small. (3) Each configuration was trained once. (4) Messages are routed one at a time, so risk expressed across several turns can be missed. (5) The LLM judge shares a model family with the generator, favours long replies and missed unsupported content; a blinded human evaluation is planned. (6) The system is English-only.

6 Conclusion

We presented a two-stage mental health support chatbot whose router is trained only on published, independently annotated datasets. A ten-configuration ablation showed that test-split scores can hide severe failures, with a fine-tuned router scoring 0.833 on the test split but 0.281 on held-out messages, and that targeted public datasets close most of this gap: the selected router reaches 0.849 held-out macro-F1, recalls 0.914 of crisis messages from unseen users at a 0.3% false-alarm rate, and confines all held-out false alarms to context-aware level-2 replies. In the generation stage, grounding rules reduced attributed emotions from 62.0% to 8.0% of replies. Future work will add an independent crisis test set, multi-turn risk tracking, human evaluation and Vietnamese-language support.

Data Availability. All datasets are publicly available from their original sources; the label-mapping rules are given in Table 3. The files used in this study are:

- MentalChat16K: https://huggingface.co/datasets/ShenLab/MentalChat16K
- C-SSRS Reddit: https://zenodo.org/records/4543776
- CLINC150 (`plus` configuration): https://huggingface.co/datasets/clinc/clinc_oos
- GoEmotions: https://github.com/google-research/google-research/tree/master/goemotions
- HWU64: https://github.com/xliuhw/NLU-Evaluation-Data
- MultiWOZ 2.2: https://github.com/budzianowski/multiwoz/tree/master/data/MultiWOZ_2.2
- ESConv: https://github.com/thu-coai/Emotional-Support-Conversation

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